D72 — The Proton Asymmetry at the Core–Sea Interface: Forced Partition $(28, 28, 24)$, Relocated Leakage Structure, and the Exact Valence Spin Share $\Delta\Sigma = 24/80$

Cédric Laubscher
cedric@cedriclaubscher.ch

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Abstract

We report the resolution of an open internal inconsistency of the leakage mechanism documented in D71 (DOI: 10.5281/zenodo.23776625): the three valence cores were modelled as three identical K_4 graphs, although the corpus records an asymmetric core link structure 28/28/24. Two hypotheses for the locus of this asymmetry were declared a priori (campaign v17): H_A, internal core relations — which is falsified, since it destroys the 918/930 site partition, the identities $144 = 12^2$, $B = 528$, $\delta = 1/56$, and is combinatorially impossible on K_4 sites — and H_B, core→sea interface relations, which preserves every integer result of the symmetric campaign and identifies the spin reference core with the short interface, the down-quark core. Under H_B the interface sum is $I_{\mathrm{tot}} = 80 = 11115 - 11035$, and within a declared family of 2+1 partitions, $(28, 28, 24)$ is the unique partition satisfying two independent structural criteria (v18-A). The wake census of D71 survives interface weighting exactly ($184.8 + 184.8 + 158.4 = 528$; v18-B1) and is re-pinned by the distinguished interface through the single hit $24 \times 22 = 528$ on the declared family $x(x-2)$ (v18-B2). The two Euler loci of v13 are recovered on the bow of the down-core only (predicted degrees $\{29, 29, 25\}$, observed $\{25, 25\}$; v18-C). The valence spin share improves from 7/24 (residual 0.167 tolerance) to exactly $\Delta\Sigma = I_d/I_{\mathrm{tot}} = 24/80 = 0.300000$ (residual 0.000; v18-D), superseding 7/24 by a frozen tie-break rule. The diagonal-cut fraction $1/(2\sqrt{2})$ admits no interface-rational competitor (v18-E), and the closed fraction $f = 144/917$ is immune to the asymmetry (v18-F). The unified triple-target audit passes with residuals 0.64σ (proton skin), 0.10σ (closed fraction) and 0.000 (spin) (v18-G). All claims retain status (T) theorem or (D) derived within a priori declared families; the circularity of G is *not* lifted.

1 Epistemic status and reading rules

Every statement carries one of three labels, as in D71:

- **(T)** theorem-level: follows from the axioms C1–C4 and integer facts of the corpus, without free parameters;
- **(D)** derived within a family declared *a priori*, under an explicit null model;
- **(H)** heuristic or compatibility-level: survives confrontation but is not uniquely selected.

The main claim of this note is at level **(D)**: the proton asymmetry 28/28/24 is located at the core–sea interface, is uniquely forced within its declared family, and improves the spin target to an exact hit. Nothing here is a derivation of G.

2 The question: where does the proton asymmetry live?

D71 modelled the three valence cores as three identical K_4 graphs (12 vertices, 18 internal relations), with asymmetry carried only by the odd sea parity ($11017 = 23 \times 479$) and by the dynamical selection of a spin reference core. The corpus, however, records a structural asymmetry 28/28/24. Before any further derivation, this inconsistency had to be resolved by determining the *locus* of the asymmetry. Two hypotheses were declared a priori:

- $\mathbf{H}_A$: 28/28/24 are the *internal* relation counts of the three cores;
- $\mathbf{H}_B$: 28/28/24 are the core→sea *interface* relation counts, the internal K_4 structure being preserved.

3 Falsified branch $\mathbf{H}_A$: internal asymmetry (v17, D-negative)

H_A requires at least 8 sites per core (24 core sites in total) and yields three independent obstructions:

- the maximum number of edges on 4 sites is $\binom{4}{2} = 6 < 24, 28$: internal counts 28/28/24 are impossible on K_4 sites;
- $V_{\text{sea}} + 24 = 918 + 24 = 942 \neq n_u = 930$; equivalently H_A would force $V_{\text{sea}} = 906 \neq 918$;
- H_A also kills $144 = 12^2$ (hence $f = 144/917$), $B = n_u(n_u - 2) = 528$, and $\delta = 1/\binom{8}{3} = 1/56$.

Verdict: $\mathbf{H}_A$ is falsified at the fixed $n_u = 930$, $V_{\text{sea}} = 918$ partition. The falsified branch is part of the evidence.

4 Viable branch $\mathbf{H}_B$: the interface asymmetry (v17)

Under H_B the site partition is preserved: $918 + 12 = 930$, and $144 = 12^2$, $f = 144/917$ survive unchanged. The interface pattern is 2+1 with a distinguished short interface (d: 24) and two identical long interfaces (u: 28) — the uud shape. Two structural consequences follow: (i) the spin reference core is no longer a gauge choice but is structurally identified with the d-core (a D-level strengthening of the v13 Euler localisation); (ii) the grand relation total becomes 11115, with 98 core-associated relations, hence $I_{\text{tot}} = 11115 - 11035 = 80$ interface relations shared across the three cores. **Verdict: $\mathbf{H}_B$ is viable and strengthens the mechanism.**

5 Forced partition $(28, 28, 24)$ (v18-A)

Within the declared family of 2+1 partitions of $I_{\text{tot}} = 80$, three candidates pass the initial screen: $(27, 27, 26)$, $(28, 28, 24)$, $(29, 29, 22)$. Two independent structural criteria — the boundary deficit per core ($= 4$) and the divisibility of the short interface by the bow degree $d_{\text{bow}} = 8$ — each select $(28, 28, 24)$ uniquely. Notably, the symmetric split 80/3 is non-integer: *the asymmetry is required* for every core to carry an integer interface. The partition is not a free parameter. Status: **(D)**.

6 The weighted wake census (v18-B1/B2)

B1. Weighting the wake by the interface shares $w = \{u_1 : 28/80,\ u_2 : 28/80,\ d : 24/80\}$ gives per-core wake shares 184.8/184.8/158.4, summing *exactly* to 528: the wake census of D71 survives the asymmetry. The fractional parts $(0.8, 0.8, 0.4)$ sum to 2.0, the number of Euler loci (recorded as (H), no claim). Status: **(D)**. **B2.** On the declared family $x(x - 2)$ over interface integers, the single hit is $24 \times 22 = 528$: the wake total is re-pinned by the distinguished interface $I_d = 24$. Status: **(D)**.

7 Localisation: the bow of the down core (v18-C)

The predicted Euler-locus degrees per core are $\{u : 29,\ u : 29,\ d : 25\}$. The v13 campaign found exactly two loci of degree 25, both on the bow of the spin-carrying core. Under H_B this identifies the bow unambiguously with the d-core — the down-quark core is the spin reference core. Status: **(D)**.

8 The exact spin share (v18-D)

The interface reading of the valence spin budget gives

$$\Delta\Sigma \;=\; \frac{I_d}{I_{\mathrm{tot}}} \;=\; \frac{24}{80} \;=\; 0.300000,$$

against the EMC/COMPASS target 0.30 ± 0.05: residual 0.000 tolerance — the first exact spin hit of the campaign. By the frozen tie-break rule (exactness plus structural reading, both declared before fitting), 24/80 *supersedes* 7/24 (residual 0.167). The form $24/80 = 3/10$, excluded as off-chain in v16 under the symmetric core hypothesis, is admitted once the interface integers are opened by v17. Status: **(D)**.

9 Robustness of the remaining closed forms (v18-E/F)

E. No interface-rational fraction (8/24, 10/28, 28/80, ...) falls inside the T1 skin window: the gap $1/(2\sqrt{2})$ retains its geometric origin as the Euler diagonal cut (the u-share $28/80 = 0.35$ sits at 5.26 tolerance — an honest negative: the asymmetry does *not* repair the T1 gap). **F.** $f = 144/917$ counts core *sites*, never interface links, and is therefore immune to the asymmetry; residual unchanged at 0.102 tolerance. Status: **(D)** throughout.

10 Unified triple-target audit (v18-G)

Target	Prediction	Observation	Residual	Verdict
T1 skin Δr_{val}	0.0479596451424	$0.04795969 \pm 6.98 \times 10^{-8}$	0.64σ	PASS
T2 closed fraction f	$144/917 = 0.1570338$	$0.157126 \pm 9 \times 10^{-4}$	0.10σ	PASS
T3 spin $\Delta\Sigma$	$24/80 = 0.300000$	0.30 ± 0.05	0.000	PASS

The full boundary count is $B_{\mathrm{full}} = 528.3714105334503$ $(528 + 1/(2\sqrt{2}) + 1/56)$. Cross-coherence: the single integer $24 = 3 \times 8$ locks simultaneously $B = 24 \times 22$ and $\Delta\Sigma = 24/80$.

11 Limitations and falsifiers

- The circularity of G is *not* lifted: the continuous chain C1–C4 $\to B \to \Delta r_{\mathrm{val}} \to \varepsilon_G \to G_{\mathrm{PDL}}$ remains to be written (OP-D71-1).
- The T1 fractional signature remains at $\sim 1.7\sigma$; the T2 hits are correlated; the EMC window is wide and the exactness of 24/80 is relative to a ± 0.05 window.
- The interface grand total 11115 (98 core-associated relations) must still be re-derived from C1–C4 rather than patched from the corpus registry.
- The pre-registered falsifier stands unchanged: any admissible boundary law must reproduce simultaneously $\Delta r_{\mathrm{val}} = 0.04795969 \pm 7 \times 10^{-8}$ and $f = 0.157126$ from a single law.

12 Reproducibility

Campaigns v17 and v18 are pre-registered and dated 2026-10-29. The two scripts (`PDL_proton_leakage_v17.py` `PDL_proton_leakage_v18.py`; standard library only, Colab-ready) and their JSON summaries are distributed together with this note. All families, windows and tie-break rules are declared in the scripts before the fits.

13 Conclusion

The proton asymmetry was not lost in D71 — it was misplaced. Located at the core–sea interface, it is forced uniquely to $(28, 28, 24)$, re-pins the wake census on the down-core interface $I_d = 24$, identifies the spin bow with the down core, and upgrades the valence spin share to the exact value $\Delta\Sigma = 24/80 = 0.3$. The skin and closed-fraction results are unchanged. The mechanism of D71 therefore survives its own structural correction and emerges strengthened, with one fewer degree of freedom and one more exact target.